Algorithms and data structures I.

Hash tables

New data structure: Hash tables

- Goal: efficient SEARCH on a set of known (unique) keys,
  with also efficient DELETION and INSERTION methods

- We already know:

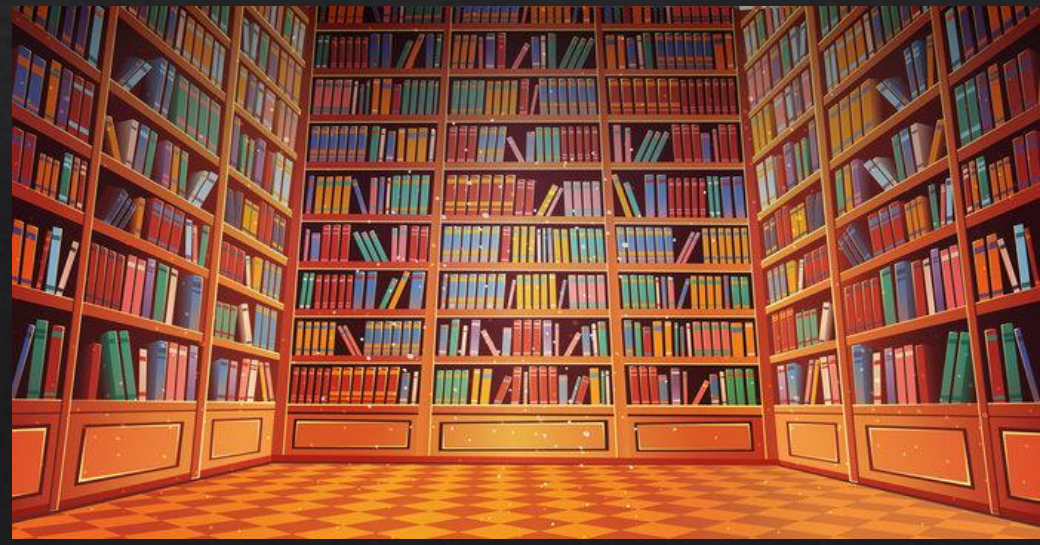
1. Unordered list/array + linear search O(n)
2. Ordered array + logarithmic (binary) search O(log n)
3. Ordered list + Efficient linear search O(n)
4. Binary searchtree: O(h) where h is the height of the tree.

- Hash: Possible to even do statistically CONSTANT time.

Illustration problem

- Let's assume you want to fill up a library with books and not just stuff them in there, but you want to be able to easily find them again when you need them.
- With the title of the book, can give you the right spot at once, so all you have to do is just stroll over to the right shelf, and pick up the book.
- But how can that be done? Well, with a bit of forethought, instead of just starting to fill up the library from one end to the other, you devise a clever little method. You take the title of the book, run it through a small computer program, which spits out a shelf number and a slot number on that shelf. This is where you place the book.
- The beauty of this program is that later on, when a person comes back in to read the book, you feed the title through the program once more, and get back the same shelf number and slot number that you were originally given, and this is where the book is located.

Source: https://stackoverflow.com/a/730813



Source: https://stock.adobe.com/search?k=library+cartoon&asset_id=266015225

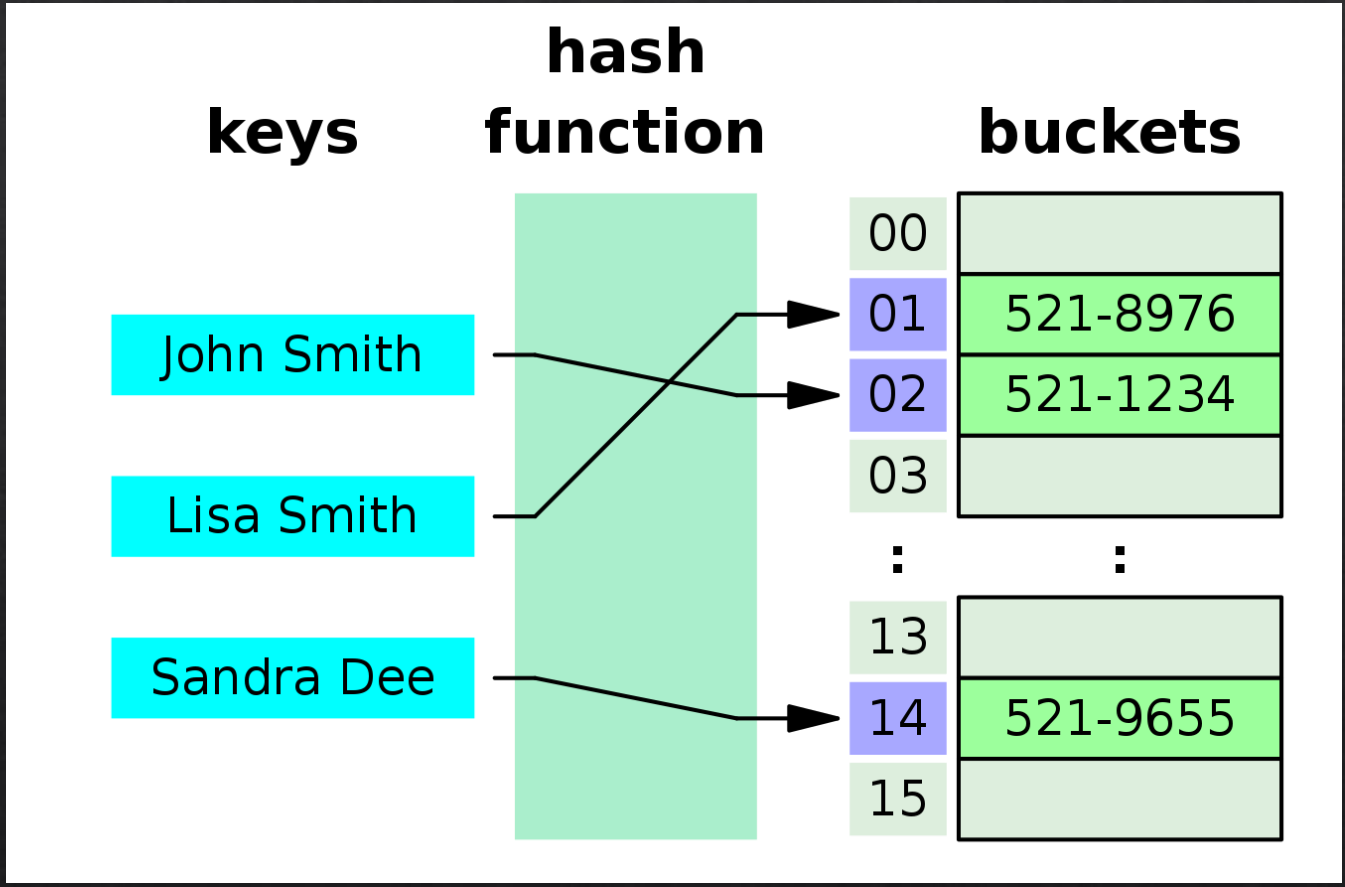
I. Direct-addressing

I. Direct-addressing

How does it work?

- Instead of inserting every new element one after the other as new neightbours
- Every element has a unique KEY (e.g. books with author + title)
- Convert the complicated KEY into a Data structure address (e.g. starting with index 0 until n, n := size-1) using the HASH function
- Put the element to this specific location.

NEW PROBLEM ?????



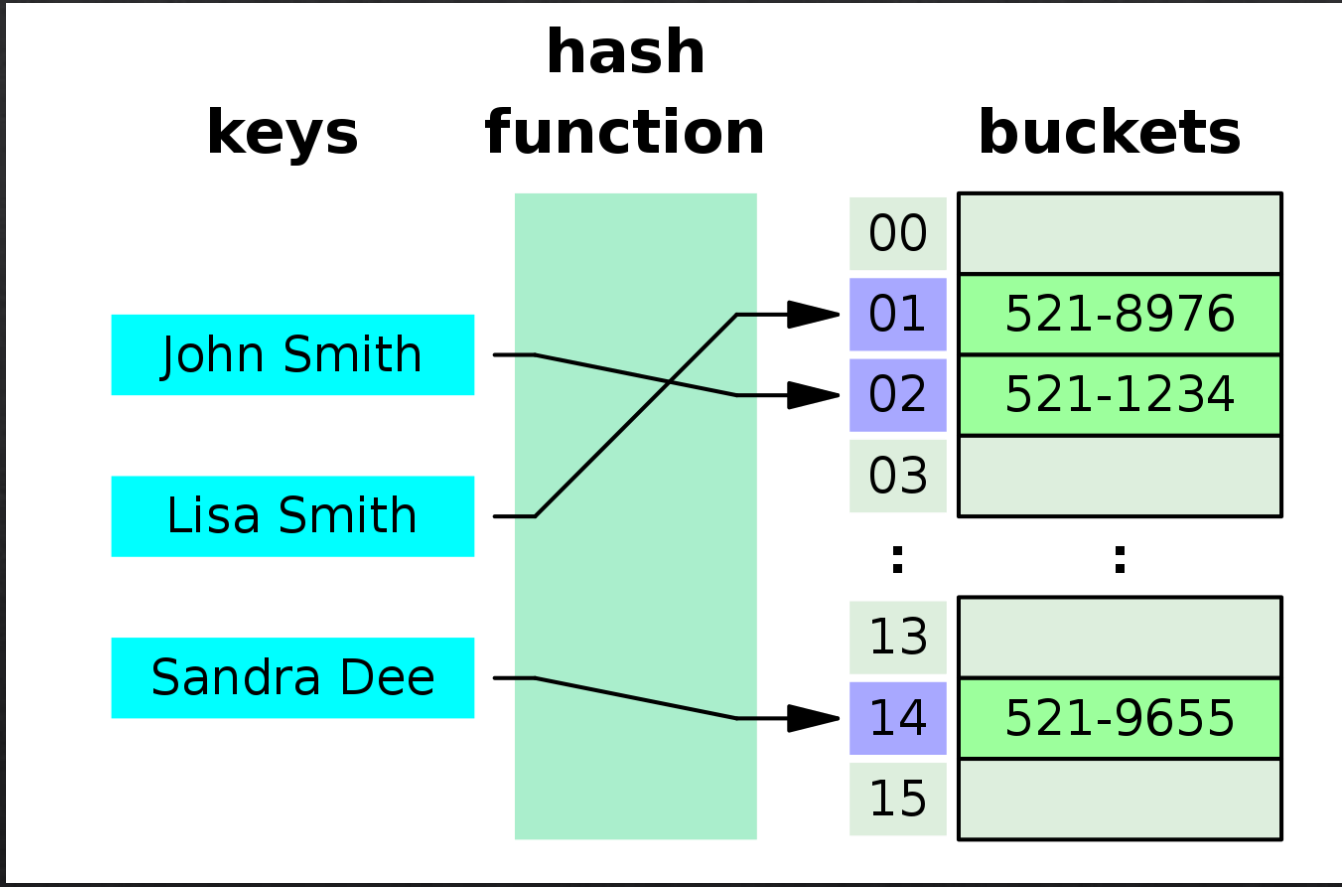
Source: https://en.wikipedia.org/wiki/Hash_table

I. Direct-addressing

What are the problems with this method?

- Space problem:
  - If I create a specific unique slot for every possible element
  - Many of wasted space if not many elements are stored from the KEY UNIVERSE
    (notation: |U| >> n)
- Possible solution:
  - Create a smaller data structure than the KEY UNIVERSE

NEW PROBLEM??????



Source: https://en.wikipedia.org/wiki/Hash_table

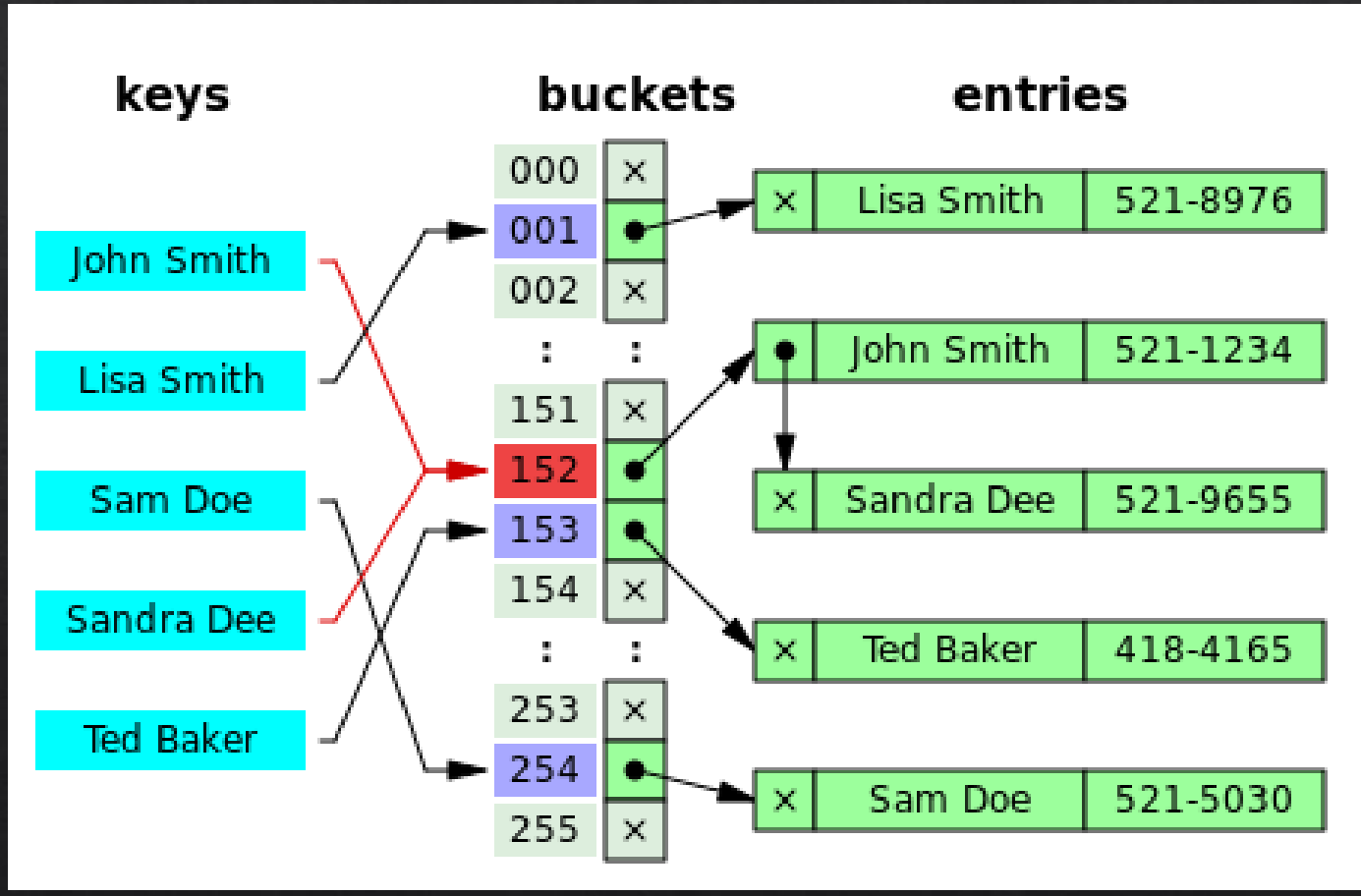
I. Direct-addressing

What are the problems with this method?

- COLLISION problem:
  - Resticting the addresses smaller than the UNIVERSE will result in mulitple keys having the same specific slot
- Possible solution:
  - Create a list at every address, and add the new elements into the list

ADVANTAGES ?????

DISADVANTAGE ?????



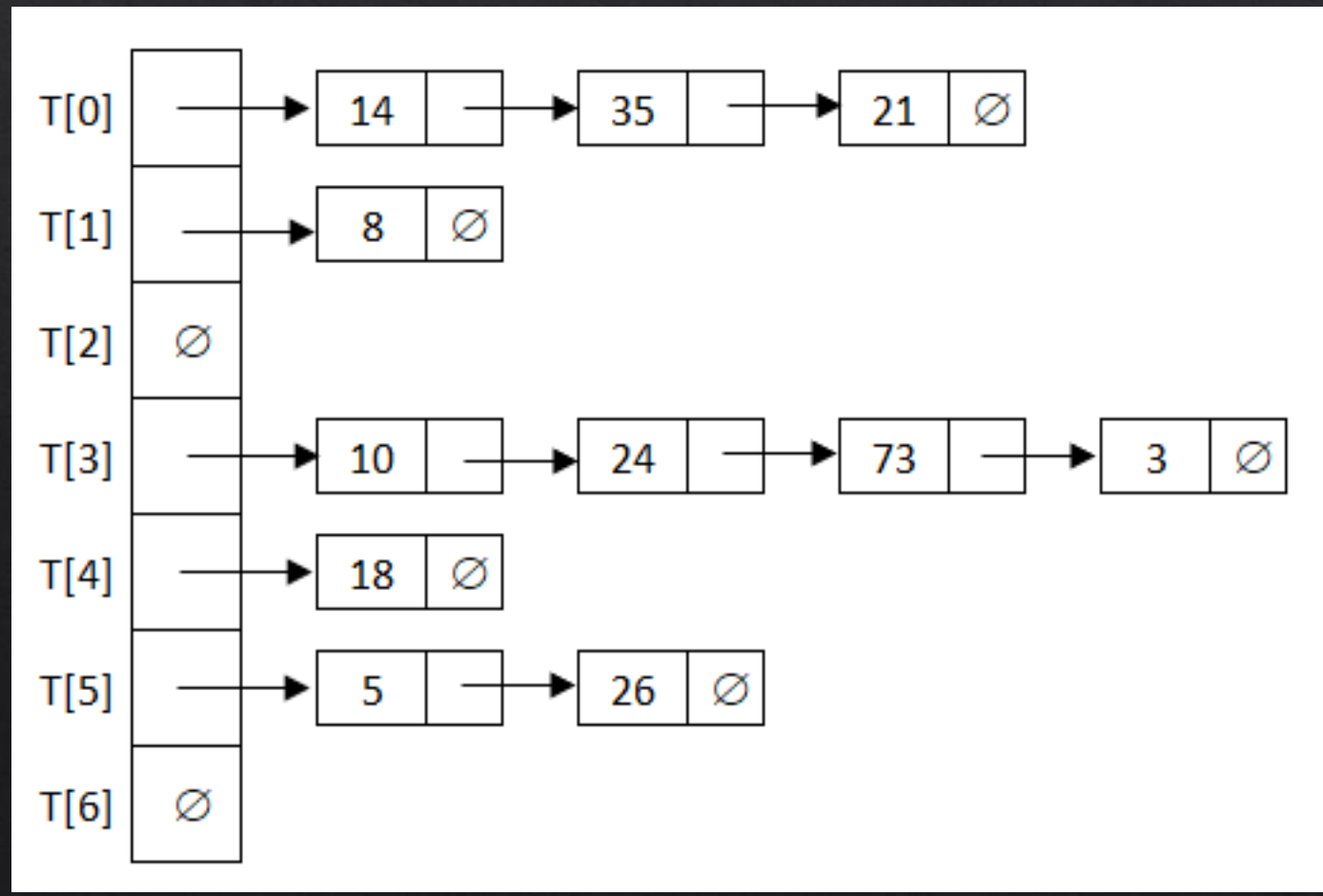
Source: https://en.wikipedia.org/wiki/Hash_table

I. Direct-addressing

Task:

- m = 7, h : N -> 0..6, h(k) = k mod 7 Insert: 3, 21, 73, 8, 24, 35, 26, 14, 5, 10, 18.
- k: key of the element
  m: size of the table
  h(k): hash function for the address

k	h(k)
3	3
21	0
73	3
8	1
24	3
35	0
26	5
14	0
5	5
10	3
18	4



Hash functions

Hash functions

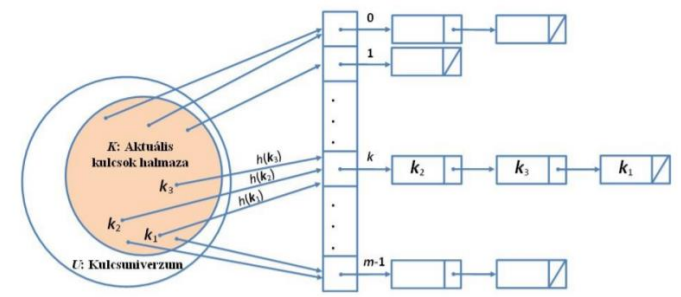
- The h : U → [0..m) Function is good if it distributes the possible keys evenly in the hash table.
- **Division methond:**
- h(k) = k mod m
- It's quality depends on m, and it's usually a good idea to make it a prime, which is far from some exponentials of 2.

Load factor of the table: $\alpha = n/m$ (n records in the table, m length of table). This gives a good approximation of how long the lists are in chained approaches, or how many key collisions it takes to successfully finish insertion.

Hash functions

- **Multiplication method:**
- Let $0 < A < 1$ be a constant (non-trivial values).
- Let's take the fraction part: $\{k*A\} < 1$
- $h(k) = \lfloor \{k*A\} * m \rfloor$ $h(k) \in [0.. m-1]$ true.
- m can be picked freely
- Fun fact: D. E. Knuth advice for A: 0.618… (1 - golden ratio???)

Chained solution



E1
$+key : U$ $\ldots$ // satellite data may come here $+next$: **E1***
$+$**E1**() { $next := \oslash$ }

init(T:E1*[m])

$i := 0$ to $m-1$

$T[i] := \oslash$

search(T:E1*[] ; k:U):E1*

return searchS1L($T[h(k)]$,k)

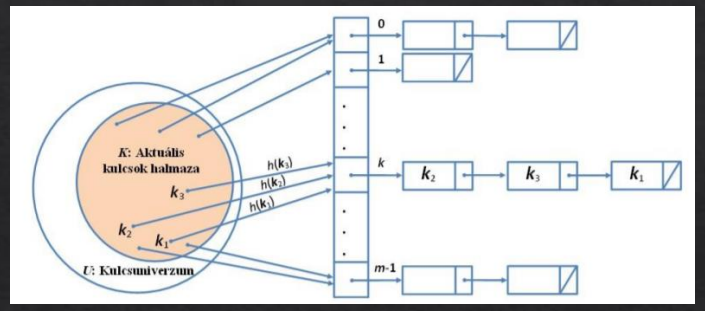
searchS1L(q:E1* ; k:U):E1*

$q \neq \oslash \wedge q \rightarrow key \neq k$

$q := q \rightarrow next$

return q

Chained solution



E1
$+key : U$
... // satellite data may come here
$+next$: **E1***
+E1() { $next := \oslash$ }

searchS1L(q:E1* ; k:U):E1*

$q \neq \oslash \wedge q \rightarrow key \neq k$

$q := q \rightarrow next$

return q

insert(T:E1*[] ; p:E1*):$\mathbb{B}$

$k := p \rightarrow key$; $s := h(k)$

searchS1L$(T[s], k) = \oslash$

$p \rightarrow next := T[s]$	return $false$
$T[s] := p$	
return $true$	

remove(T:E1*[] ; k:U):E1*

$s := h(k)$; $p := \oslash$; $q := T[s]$

$q \neq \oslash \wedge q \rightarrow key \neq k$

$p := q$; $q := q \rightarrow next$

$q \neq \oslash$

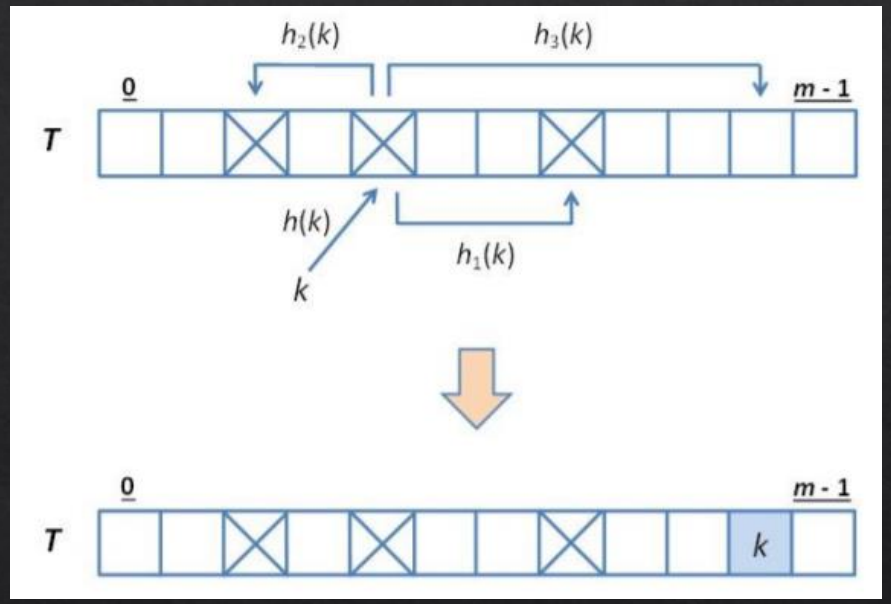
$p = \oslash$

$T[s] := q \rightarrow next$	$p \rightarrow next := q \rightarrow next$	SKIP
$q \rightarrow next := \oslash$		

return q

II. Open-addressing

- Each record goes into the table directly
- We resolve collisions by trying to find a new free slot somehow
- Potential probing sequence:
- h(k,0) h(k,1) h(k,2) … h(k,m-1)
- The probing sequence must ’touch’ every slot once, therefore is it the permutation of 0..m-1 integers
- For the same key we must get the same probing sequence



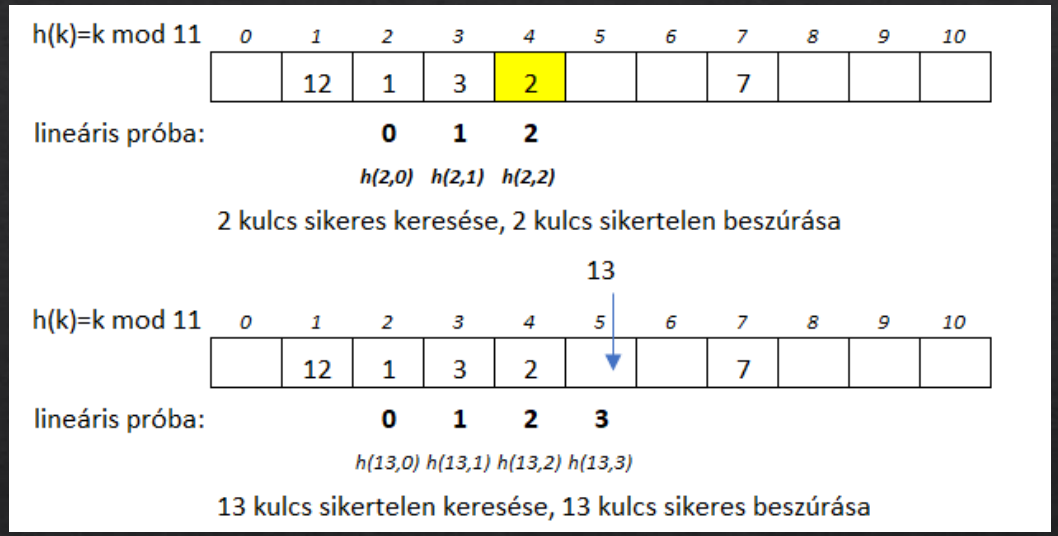
Notations

R
+ $k : U \cup \{E, D\}$ // k is a key or it is Empty or Deleted
+ ... // satellite data

- k is a key,
  E – empty, D- deleted. E,D are extremal values
- h:U x 0..(m-1) → 0..(m-1) : hashing sequence
  ⟨h(k,0),h(k,1),h(k,2), … h(k,m-1)⟩ : potential probing sequence
- We say that an E, and D slots are ’free’, otherwise occupied.
- Probing sequence:
  - **Linear, Quadratic, and double hash** *(last is well distributed)*

Open addressinga and operations with deletions.

- First consider search and insertion only
- **Search:** We follow the potential probing sequence until we find E, D or k itself. Given that we find E, it means that the key is not part of the hash table, and therefore it's not successful. If we find a D, that is the same as if the slot were occupied, since k could occur later on.
- **Insertion:** We follow the potential probling sequence until we find E, D or k itself. Given that we find E, it means that the key is not part of the hash table, and therefore we can insert it right there. If we find a D, that is the same as if the slot were occupied, however we must remember this slot since the next time we find an E, k should be inserted into the fi



Linear probing

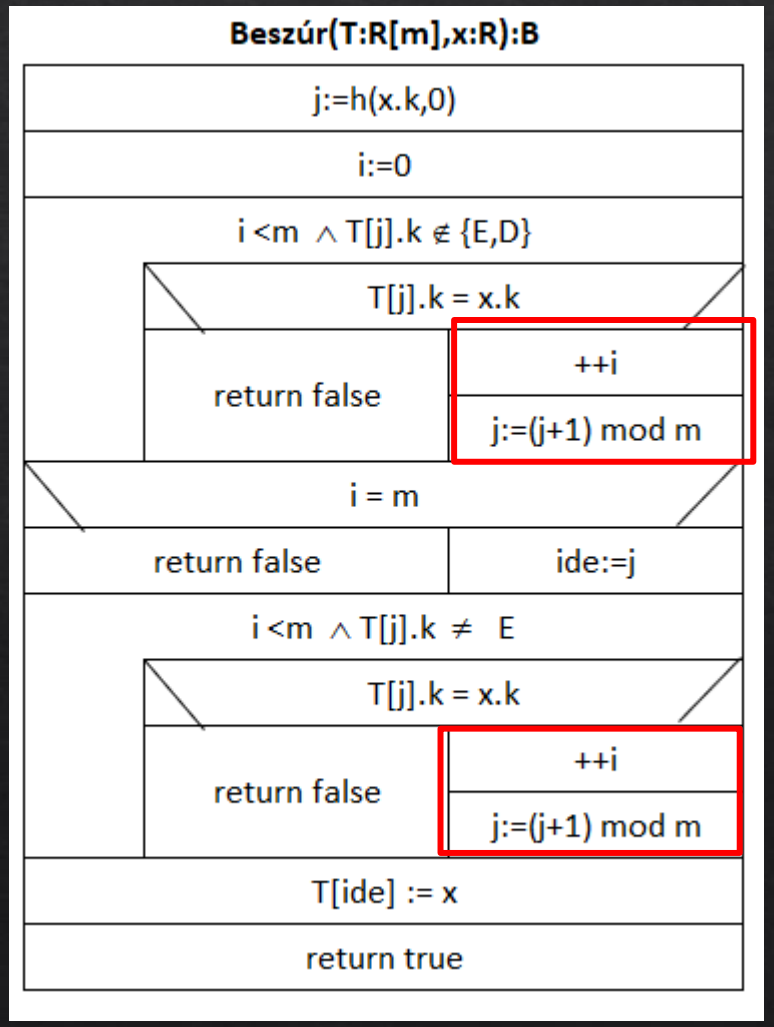
- Linear probing sequence: h(k,i) = (h(k,0) + i) mod m i=0..m-1
- Beszúr: Insert
- Töröl: Delete
- Keres: Search

Példa: $m = 11$, $h : \mathbb{N} \to 0..10$, $h(k, n) = (k + n) \bmod 11$.

Művelet	k	$h_1(k)$	Próbasorozat	0	1	2	3	4	5	6	7	8	9	10
Beszúr	24	2	2_E ✓			24								
Beszúr	16	5	5_E ✓			24			16					
Beszúr	57	2	$2, 3_E$ ✓			24	57		16					
Beszúr	32	10	10_E ✓			24	57		16					32
Beszúr	15	4	4_E ✓			24	57	15	16					32
Töröl	57	2	$2, 3_{57}$ ✓			24	D	15	16					32
Beszúr	21	10	$10, 0_E$ ✓	21		24	D	15	16					32
Keres	2	2	$2, 3_D, 4, 5, 6_E$ ✗	21		24	D	15	16					32
Beszúr	2	2	$2, 3_D, 4, 5, 6_E$ ✓	21		24	2	15	16					32
Beszúr	2	2	$2, 3_2$ ✗	21		24	2	15	16					32
Keres	21	10	$10, 0_{21}$ ✓	21		24	2	15	16					32

αα=6/11

Algorithm for insertion



Quadratic probing

- $h(k,i) =(h(k,0) + c_1 * i + c_2 * i^2) \bmod m$
- c_1 and c_2 constants ($c2 \neq != 0$).
- Given that m is an exponential of 2, $c_1=1/2$ and $c_2=1/2$ are good solutions.
- $h(k,i) = (h(k,0) + ½ * i + ½ * i^2) \bmod m$, i=0..m-1.
- $h(k,i) = (h(k,0) + 1 +2 + 3 + \ldots..+i) \bmod m$, i=0..m-1.

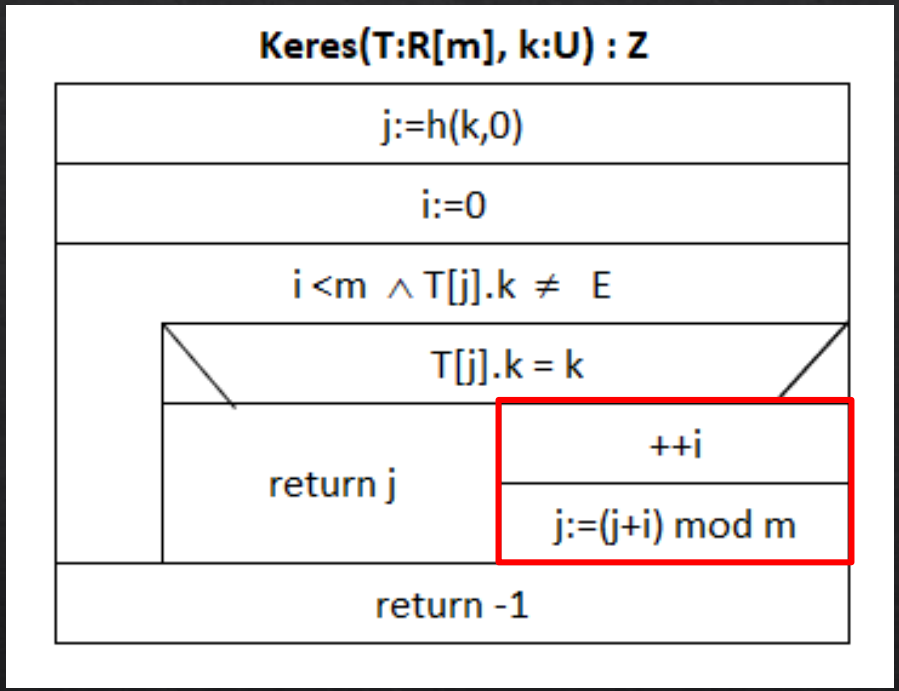
Quadratic

Példa: $m = 8$, $h : \mathbb{N} \to 0..7$, $h(k, n) = (k + \frac{n+n^2}{2}) \bmod 8$.

Művelet	k	$h_1(k)$	Próbasorozat	0	1	2	3	4	5	6	7
Beszúr	13	5	5_E ✓						13		
Beszúr	20	4	4_E ✓					20	13		
Beszúr	31	7	7_E ✓					20	13		31
Beszúr	87	7	$7, 0_E$ ✓	87				20	13		31
Beszúr	12	4	$4, 5, 7, 2_E$ ✓	87		12		20	13		31
Töröl	31	7	7_{31} ✓	87		12		20	13		D
Keres	12	4	$4, 5, 7_D, 2_{12}$ ✓	87		12		20	13		D
Keres	15	7	$7_D, 0, 2, 5, 1_E$ ✗	87		12		20	13		D
Beszúr	4	4	$4, 5, 7_D, 2, 6_E$ ✓	87		12		20	13		4
Töröl	10	2	$2, 3_E$ ✗	87		12		20	13		4
Beszúr	35	3	3_E ✓	87		12	35	20	13		4
Beszúr	10	2	$2, 3, 5, 0, 4, 1_E$ ✓	87	10	12	35	20	13		4

αα= 7/8

Search for quadratic



Double hashing

- Similar to linear hashing, but instead of stepping one each time, the step size is determined by a second hash function
- h_1: U $\rightarrow$ 0..m-1, Provides the initial place of the key
- h_2: U $\rightarrow$ 1 m-1, Provides the step size
- h(k,i) =(h_1(k) + i*h_2(k)) mod m, i=0..m-1 is the full hash function
- h_2(k) properties:
  - h_2(k) > 0
  - h_2(k) and m relative primes (m is usually prime)

Double hashing

Példa: $m = 11$, $h_1(k) = k \bmod 11, h_2(k) = 1 + (k \bmod 10)$.

Művelet	k	$h_1(k)$	$h_2(k)$	Próbasorozat	0	1	2	3	4	5	6	7	8	9	10
Beszúr	23	1	4	1_E ✓		23									
Beszúr	42	9	3	9_E ✓		23								42	
Beszúr	31	9	2	$9, 0_E$ ✓	31	23								42	
Beszúr	110	0	1	$0, 1, 2_E$ ✓	31	23	110							42	
Beszúr	55	0	6	$0, 6_E$ ✓	31	23	110				55			42	
Töröl	55	0	6	$0, 6_{55}$ ✓	31	23	110				D			42	
Keres	72	6	3	$6_D, 9, 1, 4_E$ ✗	31	23	110				D			42	
Beszúr	14	3	5	3_E ✓	31	23	110	14			D			42	
Töröl	22	0	3	$0, 3, 6_D, 9, 1, 4_E$ ✗	31	23	110	14			D			42	
Töröl	110	0	1	$0, 1, 2_{110}$ ✓	31	23	D	14			D			42	
Beszúr	13	2	4	$2_D, 6_D, 10_E$ ✓	31	23	13	14			D			42	

αα= 5/11

Search for double hashing